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The Effects of *Zataria multiflora* Boiss. (Shirazi thyme) on Nonalcoholic Fatty Liver Disease and Insulin Resistance: A Randomized Double-blind Placebo-controlled Clinical Trial

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HIGHLIGHTS

Nonalcoholic fatty liver disease (NAFLD) is the most common type of hepatic disorders worldwide and insulin resistance (IR) is one of its main risk factors. *Zataria multiflora* Boiss (Shirazi thyme) (ZM) is a medicinal herb with antioxidant and insulin sensitizer properties. This study aimed to evaluate the effects of ZM supplementation on NAFLD and IR. ZM supplementation resulted in a significant reduction in serum insulin levels ($P=0.030$), insulin resistance ($P=0.023$), systolic ($P=0.002$) and diastolic ($P=0.009$) blood pressure in comparison with the placebo group.

Abstract

Objective: The aim of the study is to evaluate the effects of *Zataria multiflora* Boiss. (Shirazi thyme) (ZM) supplementation on Nonalcoholic Fatty Liver Disease (NAFLD) and related insulin resistance (IR).

Design: In this randomized double-blind placebo-controlled clinical trial, 85 patients with NAFLD were administered either 700 mg ZM powder or placebo twice daily for 12 weeks. All patients were advised to follow the recommendations for diet modification.

Results: ZM supplementation resulted in a significant reduction in serum insulin level (-2.72 ± 0.80 vs -0.67 ± 0.90 , $P=0.030$), insulin resistance (-0.80 ± 0.22 vs 0.06 ± 0.22 , $P=0.023$), systolic (-3.44 ± 1.05 vs 1.03 ± 0.98 , $P=0.002$) and diastolic (-2.33 ± 0.77 vs 0.26 ± 0.63 , $P=0.009$) blood pressure in patients with NAFLD in comparison with the placebo group. There was no significant difference between two groups regarding serum levels of alanine aminotransferase

(ALT), high sensitive C-reactive protein (hs-CRP), Tumor necrosis factor- α (TNF- α), grade of fatty liver in ultrasonography, lipid profiles, and other outcomes.

Conclusion: ZM supplementation with daily dose of 1400 mg for 12 weeks improved insulin resistance in patients with NAFLD. Further studies with longer duration and larger sample size are recommended. (*ClinicalTrials.gov ID: NCT02983669*)

Keywords: Zataria multiflora; Thymus plant; Nonalcoholic Fatty Liver Disease; Insulin resistance; Herbal Medicine; Dietary supplements; Traditional Medicine

1. Introduction

Non alcoholic fatty liver disease (NAFLD) is a fatty liver, when hepatic steatosis is not originated from any secondary cause of hepatic fat accumulation.¹ This is the most common type of hepatic disorders worldwide.² Metabolic syndrome, central obesity, dyslipidemia, and type2 diabetes mellitus are major risk factors for NAFLD.³ Its pathogenesis is not fully understood; however, insulin resistance has been mentioned to play a key role in the pathophysiology of hepatic steatosis (and probably steatohepatitis).⁴ The disease might progress to various stages of liver damage, such as fibrosis, cirrhosis, and hepatocellular carcinoma.⁵ Other than lifestyle modifications, no definite treatment strategy has yet been approved for NAFLD.⁶ However, previous investigations have revealed that antioxidants and anti-inflammatory agents could be effective in the treatment of this disease.⁷

Complementary and Alternative Medicine (CAM) that have been recently substituted by the term “Integrative Medicine”, has received a lot of interest in the past decades.⁸ Herbal therapy is a popular branch of this medicine with a long lasting history.⁹ *Zataria multiflora* Boiss. (Lamiaceae) is an herbal plant cultivated in Iran, Pakistan and Afghanistan.¹⁰ This plant with the vernacular name ,*Avishan-e-Shirazi*, has chemical and pharmacological similarities to *Thymus vulgaris*, a well-known medicinal plant.¹¹ Persian medicine scholars, such as Rhazes (865-925 AD)¹² and Avicenna (980-1037 AD)¹³ believed that *Zataria multiflora* Boiss. (ZM, Shirazi thyme) with the traditional name, *Saatar*, cleans the liver from excessive moistures.¹⁴ Its aerial parts are not only used as condiment, but are also consumed in folk medicine for its antidiarrheal, carminative, antiseptic, anthelmintic, and analgesic properties.¹⁵ Its essential oil contains significant amounts of phenols and flavonoids, while the main components are thymol and carvacrol. *In vivo* and *in vitro* studies showed the antioxidative, anti-inflammatory, anti-diabetic, hepatoprotective and insulin sensitivity enhancing effects of ZM.^{10,16} Considering the aforementioned beneficial properties and lack of human studies, the present clinical trial was designed to evaluate the effect of ZM on NAFLD and its related insulin resistance.

2. Materials and methods

2.1. Study design

In this randomized double-blind placebo-controlled clinical trial, two parallel groups of patients with NAFLD participated from February to March, 2017 and the follow-up period ended in June, 2017. Patients who met the inclusion criteria were enrolled after signing a written informed consent. Patients were randomly allocated into treatment and placebo groups by block

randomization which was generated by a computer as a non-stratified list. Both the participants and investigators were blind to the intervention and control groups. Patients in the treatment group received two capsules (each containing 350 mg ZM) before breakfast and two capsules at bedtime for a period of 12 weeks. In the control group, patients received identical placebo (wheat flour) capsules in the same order. Patients were advised to eliminate fast foods and soft drinks from their diets. The study protocol was approved by the Local Medical Ethics Committee of Shiraz University of Medical Sciences (reference number: IR.SUMS.REC.1394.199). The study was registered in the Clinical Trials Registry (ClinicalTrials.gov ID: NCT02983669).

2.2. Subjects

A total of ninety patients (from Shahid Motahari and Faghihi outpatient clinics-Shiraz University of Medical Sciences) participated in this study.

Inclusion criteria: age between 18 and 65 years, body mass index (BMI) between 18 and 35 kg/m², evidence of fatty liver according to ultrasonography, alanine aminotransferase (ALT) equal or more than 1.5 times the upper limit of normal range (≥ 45 IU/L in male and ≥ 30 IU/L in female).

Exclusion criteria: pregnancy, lactation, alcohol consumption, diabetes mellitus, history of sensitivity to ZM or thyme, acute or chronic liver disease, malignancy, usage of hepatotoxic drugs within the past 6 months, usage of drugs with effect on the biochemical tests of the study within the previous 3 months (e.g., metformin, vitamin E, oral contraceptive pills, omega-3, statins, glucocorticoids), hypo and hyperthyroidism, renal insufficiency (serum creatinine ≥ 1.5 mg/dl), and unwilling to continue participation.

2.3. Measurements

At the beginning and the end of study, all participants were evaluated according to blood pressure, height, weight, and waist and hip circumferences. A stadiometer and a standard weighing balance was applied for height and weight measurement, respectively. The BMI was calculated with the following formula: $BMI = \text{Kg}(\text{Weight})/\text{m}^2(\text{Height})$. A standard flexible tape was used to measure waist and hip circumferences in standing position. In addition, a standard mercury sphygmomanometer was used for double measurement of blood pressure on the right arm in the sitting position after a 15-minutes rest. It is worth mentioning that all measurements were taken by one trained person before and after intervention.

Blood samples were taken from all patients after 12 hours of fasting. Biochemical tests were measured at weeks 0 and 12, except complete blood count (CBC), serum creatinine, alanineaminotransferase (ALT) and aspartate aminotransferase (AST) which were measured at weeks 0, 6 and 12. Photometric assay (Pars Azmoun Company) for measurement of ALT and AST, and enzymatic (glucose oxidase) colorimetric method by means of standard kits (Pars Azmoun Company, Iran) for measurement of blood glucose level (FBS) were used.

Standard kits (Izotop Company, Hungary) for immunoradiometric assay (IRMA) method were used to assess serum insulin levels ($\mu\text{IU}/\text{ml}$). The following equation of homeostasis model of assessment index (HOMA-IR) was used to assess insulin resistance: $\text{Fasting insulin } (\mu\text{IU}/\text{ml}) \times \text{FBS } (\text{mg}/\text{dl})/405$.¹⁷

Bionic Company Standard kits were used in enzymatic colorimetric method to determine the lipid profile comprising of LDL-cholesterol, HDL-cholesterol, total cholesterol, and triglyceride. Gamma-Glutamyl Transferase (GGT) was measured using an enzymatic colorimetric assay (Pars azmoun). High sensitivity CRP (C-reactive protein) (hs-CRP) was measured using enzyme linked immunosorbent assay (ELISA) Kit (Diagnostic Biochem Canada), and the concentration

of Tumor necrosis factor- α (TNF- α) was measured using an ELIZA Kit (Bioassay Technology Laboratory).

Additionally, liver ultrasonography was performed by the same radiologist using Philips IU22 machine. Fatty liver was scored according to its echogenicity into the following 3 grades:

1. Mild increased hepatic echogenicity with normal visualization of the diaphragm and intrahepatic vessels.
2. Moderate increased hepatic echogenicity with slightly impaired visualization of the diaphragm and intrahepatic vessels.
3. Severe increased hepatic echogenicity with poor or non-visualization of the diaphragm and intrahepatic vessels.

Ultrasonography was performed in Faghihi hospital clinic at weeks 0 and 12.

2.4. Volatile composition of ZM and Preparation of drug and placebo capsules

Three samples of ZM (aerial parts) were purchased from three different medicinal plants store and were separately authenticated by a botanist from the Department of Phytopharmaceuticals, School of Pharmacy, Shiraz University of Medical Sciences, Shiraz, Iran. Each sample was specified with a voucher number. Samples were then subjected to a hydrodistillation process using a Clevenger apparatus. Through this process, the amount of yielded essential oil from each sample was considered as a parameter for selecting the best sample for further steps.

Afterward, yielded essential oils were subjected to Gas Chromatography/Mass Spectrometry (GC/MS). This procedure was carried out to identify the major constituents in essential oil samples as well as all volatile compounds. The GC/MS procedure was performed using a gas chromatography (Agilent technologies-7890A) with a HP-5MS capillary column (phenyl methyl

siloxane, 30 m × 0.25 mm i.d.), coupled to a mass detector (Agilent technologies model-5975C). Helium as a carrier gas was adjusted to a flow rate of 1 ml/min and a split ratio of 1:30. The injector and detector temperature values were adjusted to 250 and 280°C, respectively. Column temperature was programmed linearly from 60 to 250°C (ramp ~5°/min) and then was held at 250°C (15 min). The interface temperature was set to 280°C. Mass spectrometer was acquired at a mass range of 30-600 m/z, in EI mode (70 eV). ZM essential oil was diluted in dichloromethane (~1%) and subsequently injected into the system. A homologous series of n-alkanes (C₈-C₂₂) were also subjected to GC/MS. Identification of constituents was done by determining the Kovats indices (KI), and by comparing the derived KI and mass spectra data with the findings of previous studies.

Each drug capsule was filled with 350mg ZM aerial part cleaned powder (sieved with mesh ≈ 35). In parallel, placebo capsules were filled with 350mg of wheat flour (heated until it turned brown), including 35mg of ZM aerial part powder (to improve blindness regarding taste and aroma). Both placebo and ZM capsules were packaged in similar containers and labeled accordingly.

2.5. Statistical Analysis

Considering a similar study on Ginger, the maximum sample size was calculated for HOMA-IR (Mean±SD in placebo group was 2.38±0.2 and in intervention group was 2.18±0.3).¹⁸ A sample size of 52 (26 persons in each group) was calculated, considering a power of 90% and $\alpha=0.05$. With the expectation of potential loss to follow-up, 90 patients (45 persons in each group) were enrolled. SPSS software, version 22 was used for data analysis. Baseline characteristics were compared between the two groups by using independent sample *t*-test. Wilcoxon and Mann-Whitney tests were used to compare changes in the variables within and between groups,

respectively. To eliminate the effects of confounding factors, ANCOVA test was used and adjusted for age, sex, weight, ALT, AST, hip circumference and basal value of each variable. $P < 0.05$ was considered significant.

3. Results

3.1. Composition of ZM (Shirazi thyme) aerial part

In this study, from three ZM samples, a sample with more than 2% (V/W) of essential oil was selected for further steps. GC/MS analysis revealed that thymol, as a phenolic compound of oxygenated monoterpenes, was the main component. This compound might be responsible for the various pharmacological activities. Carvacrol, p -cymene and linalool were also obtained as the main ingredients in ZM samples.

3.2. Baseline characteristics

Eighty-five patients (45 patients in Thyme group, 40 patients in placebo group) were enrolled in the final analysis. Five patients in the placebo group (two patients due to immigration, two patients due to usage of some drugs and one patient due to missed follow-up) were excluded from the study.

The compliance rate of patients in the two groups was more than 97%. Baseline characteristics are summarized in Table 1. There were no significant differences in age, gender, BMI, waist circumference, systolic and diastolic BP, FBS, insulin level, HOMA-IR and grade of NAFLD in ultrasonography, but significant differences were observed in weight, hip circumference, ALT and AST levels between the two groups at the baseline.

3.3. Primary outcomes

There was significant reduction ($P < 0.001$) in serum ALT levels in both groups at 6th and 12th weeks of trial, the difference was more significant ($P < 0.05$) in Thyme group than placebo group (Table 2). After adjustment for the confounding factors, the difference was not significant (Table 3).

There was significant reduction in insulin resistance index (HOMA- IR) and insulin level ($P = 0.007$ and 0.048 respectively) only in Thyme group compared with placebo group, and this significance remained even after adjustment for confounding factors ($P = 0.023$ and 0.030 respectively).

Grade of fatty liver in ultrasonography was significantly reduced ($P < 0.001$) in both groups, but no significant difference was observed between the groups ($P = 0.967$).

3.4. Secondary outcomes

Decrease in anthropometric factors including weight, BMI, waist circumference, hip circumference and waist-to-hip ratio was significant in both groups at the end of the study, but the difference between groups was not significant. There was significant decline in systolic and diastolic blood pressures only in Thyme group compared with placebo group, and this significance remained even after adjusting for the confounding factors ($P = 0.002$ and 0.009 respectively). Serum levels of AST decreased significantly at week 6 and 12 in both groups, and this significance was more at week 12 ($P = 0.045$) in Thyme group in comparison with the placebo group, but after adjustment, the difference was not significant. Reduction of GGT levels was significant only in Thyme group, but the changes were not significant between groups. There was no significant change in lipid profiles, TNF- α , and hs-CRP within and between groups.

3.5. Safety measures and adverse events

CBC, platelet and serum creatinine levels at week 0, 6 and 12 were normal and there were no significant changes in both groups. There was no report of severe adverse events in any of the patients, but two patients (one patient in each group) complained of constipation which was resolved with simple recommendation.

4. Discussion

To the best of our knowledge, this is the first clinical trial investigating the effect of ZM supplementation on NAFLD characteristics. In this double-blind clinical trial, daily consumption of 1.4grams (700mg bid) of ZM aerial parts powder for 12 weeks resulted in a significant reduction in insulin resistance, serum insulin levels and blood pressure.

Insulin resistance is one of the most important risk factors in the pathogenesis of NAFLD. Mohammadi et al., showed the beneficial effect of ZM on insulin resistance in fructose fed insulin resistant rats. Their study showed that intragastric injection of ZM water extract for two weeks resulted in a significant reduction in blood glucose, insulin level, insulin resistance and serum triglyceride compared with the control group. The authors attributed these results to the increase in peroxisome proliferator activated receptor (PPAR- γ) protein expression and increase in adiponectin.¹⁰ PPAR- γ activation increases the sensitivity of insulin receptors¹⁹ and adiponectin inhibits gluconeogenesis in the liver.²⁰ *In vitro* study by Gholamhosseinian et al. showed the strong inhibitory effect of water extract of ZM on alpha glucosidase,²¹ which is one of the enteric enzymes for the digestion of complex polysaccharides.²² In another study, this extract significantly reduced postprandial glucose in diabetic rats after receiving 2g/kg maltose, but this effect was not evident in non-diabetic rats.²³

Previous studies demonstrated the hepato-protective effects of ZM. Shokrzadeh et al. showed that pretreatment with ethanol extract aerial parts of ZM for 7 days before cyclophosphamide (CP) injection inhibited CP hepatotoxicity in mice and significantly decreased serum ALT, AST and Alkaline Phosphatase (Alp) correlated with histopathological changes in the liver. The authors attributed this effect to the antioxidant activity of phenolic compounds in ZM.²⁴ A similar study showed that pretreatment with methanol extract of ZM had prophylactic effect on halothane hepatotoxicity which significantly decreased the serum ALT and Alp levels.²⁵

In addition, previous studies have shown the association between NAFLD and cardiovascular disease. Due to the role of NAFLD in the development of subsequent hypertension, it is imperative to maintain the blood pressure in normal range.²⁶⁻²⁸ However, there is no study in literature about the hypotensive effect of ZM, but previous studies have shown the effect of *Thymus vulgaris* (which has chemical and pharmacological similarities to ZM) in animal models. Osama et al. showed that the aqueous extract of *T.vulgaris* have significant antihypertensive effects on rats correlated with histopathological changes which might be due to the direct smooth muscle relaxant and vasorelaxant properties.²⁹ Aguila et al., also reported antihypertensive property of *T.vulgaris*, since it suppresses the vasoconstrictor reactivity of noradrenalin and renin-angiotensin-aldosterone system.³⁰

The results of our study showed the effects of ZM in reducing insulin resistance, serum insulin levels and blood pressure. Our patients were non-diabetic and reduction in insulin level without significant change in blood glucose might have been due to improvement in insulin activity. Safe dosage of powder of *Thymus vulgaris* is 1 to 4 grams bid.³¹ Nonsignificant reduction in other variables might have been due to low dosage of supplementation in our study, and higher dosage or other types of preparation or longer treatment time might be required.

This study had several limitations. Firstly, the lack of previous human studies, secondly the short duration of treatment, thirdly the lack of assessment of daily calorie intake of patients.

5. Conclusion

This randomized controlled clinical trial showed that consumption of ZM aerial parts powder with daily dose of 1400mg for 12 weeks improved insulin resistance and reduced blood pressure in patients with NAFLD, however, further studies with larger sample size and longer duration are recommended.

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Conflict of interest

The authors declare that they have no relevant financial interests.

Table 1. Baseline characteristics of patients with nonalcoholic fatty liver disease (N=85)

Variables*	Placebo group n=40	Shirazi thymegroup n=45	<i>P value</i>
Age (y)	41±11.03	36.8±9.20	0.056
Gender(M/F)	25/15	36/9	0.074
Weight (kg)	78.8±12.4	84.9±11.5	0.021
Height (m)	1.68 ± 0.11	1.72 ± 0.07	0.061
BMI (kg/m²)	27.9 ± 3.2	28.7 ± 3.5	0.243
Waist circumference (cm)	98.8 ± 9.1	102.1 ± 8.5	0.094
Hip circumference (cm)	103.9 ± 5.8	107.7 ± 6.1	0.006
WHR	0.95±0.05	0.94±0.05	0.605
Systolic BP(mmHg)	115.9 ± 9.6	116.9 ± 13.6	0.704

Diastolic BP(mmHg)	76.8± 6.9	78.9 ±9.2	0.247
ALT(IU/L)	73.5± 33.4	89.8 ±35.6	0.033
AST(IU/L)	39.8 ± 13	48.5 ± 17.5	0.011
FBS(mg/dl)	106.7 ± 12.6	105.2 ±9.3	0.547
Insulin(μIU/ml)	14.2± 7.1	14.8 ± 6.6	0.679
HOMA-IR	3.5± 1.8	3.9 ± 1.8	0.302
Ultrasound Grade	1.9 ±0.7	2.0 ± 0.6	0.734
GGT(IU/L)	47.5 ±19.8	50.8 ± 24.5	0.516

*: Mean ± SD

BMI, body mass index; WHR, waist-to hip ratio; ALT, alanine aminotransferase; AST, aspartate aminotransferase;

FBS,fasting blood sugar; HOMA-IR, homeostasis model formula for insulin resistance; GGT, Gamma-Glutamyl Transferase

Table 2: Mean changes from baseline in Thyme and placebo groups (N=85)

Variables*	Placebo group n=40	P value¹	Shirazi thyme group n=45	P value²	P value³
ALT(IU/L)					
Week 6	-17.0 ± 3.2	< 0.001	-30.7 ± 4.4	< 0.001	0.032
Week 12	-22.9 ± 3.3	< 0.001	-36.7 ± 4.6	< 0.001	0.038
AST(IU/L)					
Week 6	-7.1 ± 1.6	< 0.001	-14.2 ± 2.4	< 0.001	0.135
Week 12	-9.1 ± 1.5	< 0.001	-16.4 ± 2.7	< 0.001	0.045
FBS(mg/dl)					
Week 12	0.67 ± 1.6	0.492	-1.68 ± 1.29	0.248	0.142
Insulin(μIU/ml)					
Week 12	-0.67 ± 0.90	0.547	-2.72 ± 0.80	0.003	0.048
HOMA- IR					
Week 12	0.06 ± 0.22	0.302	-0.80 ± 0.22	0.003	0.007
GGT(IU/L)					

Week 12	-1.87 ± 1.84	0.135	-8.67 ± 2.90	0.001	0.097
Systolic BP(mmHg)					
Week 12	1.03 ± 0.98	0.358	-3.44 ± 1.05	0.012	0.007
DiastolicBP(mmHg)					
Week 12	0.26 ± 0.63	0.670	-2.33 ± 0.77	0.006	0.018
Ultrasound grade					
Week 12	-0.72 ± 0.12	< 0.001	-0.73 ± 0.09	< 0.001	0.967
Weight(Kg)					
Week 12	-1.47 ± 0.36	< 0.001	-2.22 ± 0.43	< 0.001	0.230
BMI(Kg/m²)					
Week 12	-0.53 ± 0.13	0.001	-0.76 ± 0.15	< 0.001	0.311
Waistcircumference(cm)					
Week 12	-2.66 ± 0.48	< 0.001	-2.57 ± 0.52	< 0.001	0.657
Hip circumference(cm)					
Week 12	-1.28 ± 0.29	0.007	-1.80 ± 0.43	< 0.001	0.284

*: Mean $\pm$ SE

1 and 2: P values for comparing the changes within groups (Wilcoxon test)

3: P value for comparing the differences between groups (Mann-Whitney test)

WHR, waist-to hip ratio; ALT, alanine aminotransferase; AST, aspartate aminotransferase;

FBS, fasting blood sugar; HOMA-IR, homeostasis model formula for insulin resistance; GGT, Gamma-Glutamyl Transferase

Table 3. Adjusted Mean changes and Standard Error of the outcomes

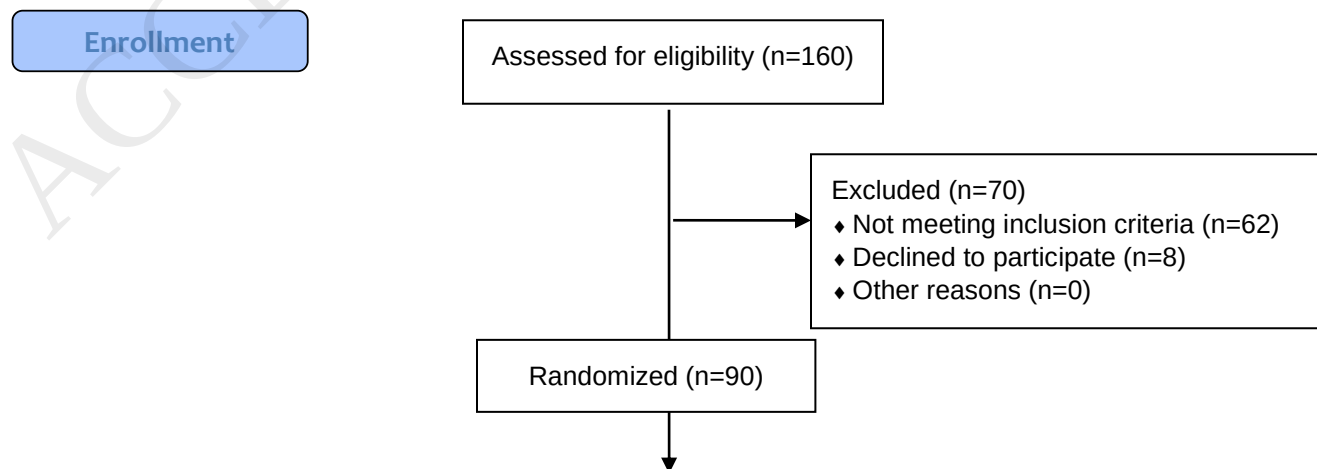
Variables	Placebo group	Shirazi thyme group	P value*
ALT(IU/L)			
Week 6	-17.0 ± 3.2	-30.7 ± 4.4	0.127
Week 12	-22.9 ± 3.3	-36.7 ± 4.6	0.344
AST(IU/L)			
Week 12	-9.1 ± 1.5	-16.4 ± 2.7	0.948
Insulin(μIU/ml)			
Week 12	-0.67 ± 0.90	-2.72 ± 0.80	0.030

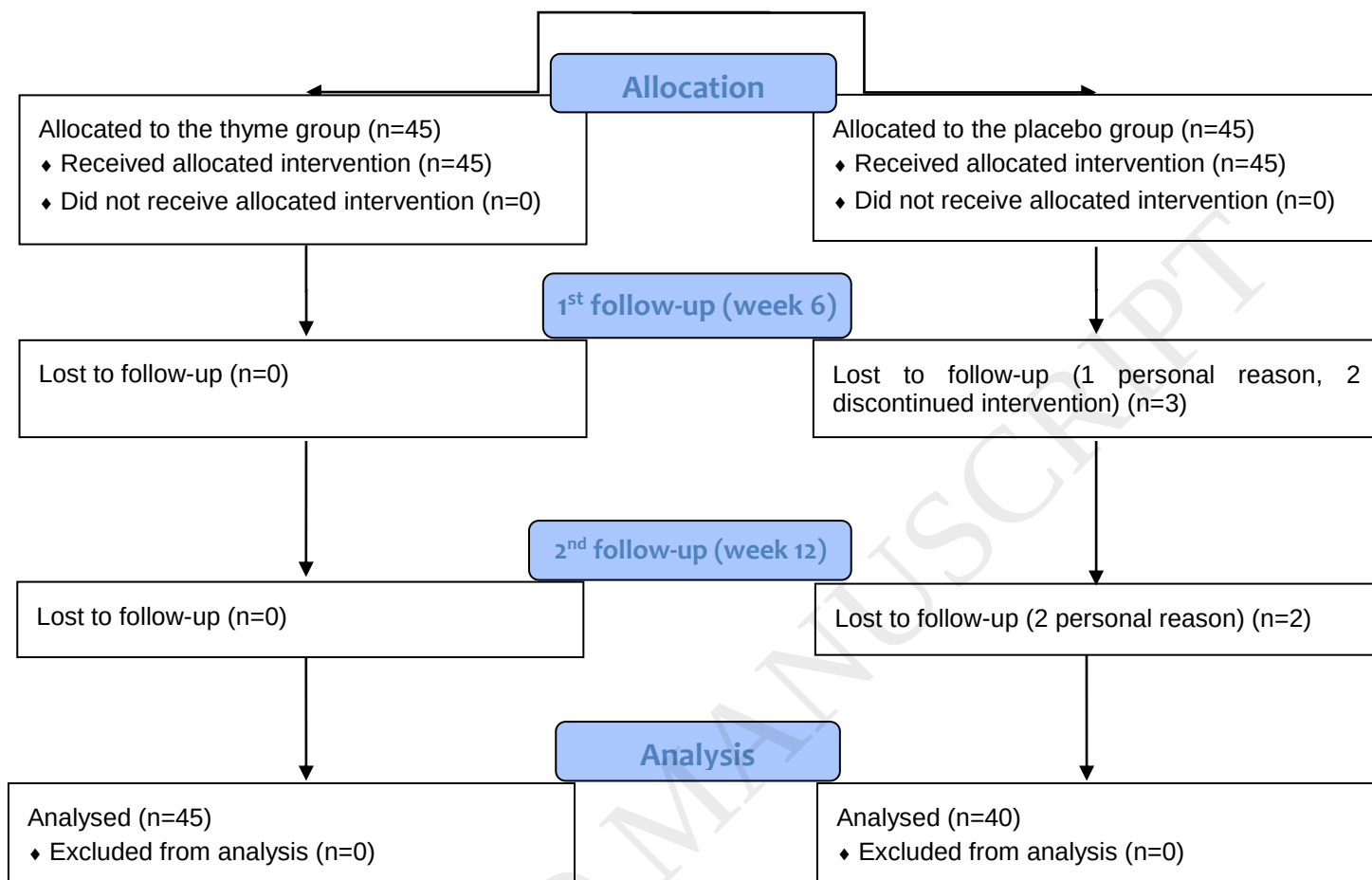
HOMA – IR			
Week 12	0.06 ± 0.22	-0.80 ± 0.22	0.023
Systolic BP(mmHg)			
Week 12	1.03 ± 0.98	-3.44 ± 1.05	0.002
DiastolicBP(mmHg)			
Week 12	0.26 ± 0.63	-2.33 ± 0.77	0.009

ALT, alanine aminotransferase; AST, aspartate aminotransferase;HOMA-IR,homeostasis model formula for insulin resistance;

**P* value for comparing the differences between groups after adjustment. (ANCOVA)

Figure 1- The CONSORT flowchart of the study





References

1. Chalasani, N, Younossi, Z, Lavine, JE, Diehl, AM, Brunt, EM, Cusi, K, et al. 2012. The diagnosis and management of non-alcoholic fatty liver disease: Practice Guideline by the American Association for the Study of Liver Diseases, American College of Gastroenterology, and the American Gastroenterological Association. *Hepatology*55(6): 2005-2023.
2. Masarone, M, Federico, A, Abenavoli, L, Loguercio, C, Persico, M. 2014. Non alcoholic fatty liver: epidemiology and natural history. *Reviews on recent clinical trials*9(3): 126-133.
3. Younossi, ZM, Stepanova, M, Afendy, M, Fang, Y, Younossi, Y, Mir, H, et al. 2011. Changes in the prevalence of the most common causes of chronic liver diseases in the United States from 1988 to 2008. *Clinical Gastroenterology and Hepatology*9(6): 524-530. e521.
4. Utzschneider, KM, Kahn, SE. 2006. The role of insulin resistance in nonalcoholic fatty liver disease. *The Journal of Clinical Endocrinology & Metabolism*91(12): 4753-4761.
5. Wong, VS, Chan, HY, Hui, A, Chan, KF, Liew, CT, Chan, FL, et al. 2004. Clinical and histological features of non-alcoholic fatty liver disease in Hong Kong Chinese. *Alimentary pharmacology & therapeutics*20(1): 45-49.

6. Promrat, K, Kleiner, DE, Niemeier, HM, Jackvony, E, Kearns, M, Wands, JR, et al. 2010. Randomized controlled trial testing the effects of weight loss on nonalcoholic steatohepatitis. *Hepatology***51**(1): 121-129.
7. Eslamparast, T, Eghtesad, S, Poustchi, H, Hekmatdoost, A. 2015. Recent advances in dietary supplementation, in treating non-alcoholic fatty liver disease. *World journal of hepatology***7**(2): 204-212
8. Daneshfard, B, Sanaye, MR, Nimrouzi, M. 2018. Prolegomena to a True Integrative Medical Paradigm. *Alternative therapies in health and medicine*.
9. Jabbari, M, Daneshfard, B, Emtiazy, M, Khiveh, A, Hashempur, MH. 2017. Biological Effects and Clinical Applications of Dwarf Elder (*Sambucus ebulus* L): A Review. *Journal of evidence-based complementary & alternative medicine***22**(4): 996-1001.
10. Mohammadi, A, Gholamhoseinian, A, Fallah, H. 2014. Zataria multiflora increases insulin sensitivity and PPAR γ gene expression in high fructose fed insulin resistant rats. *Iranian journal of basic medical sciences***17**(4): 263-270
11. Simbar, M, Azarbad, Z, Mojab, F, Majd, HA. 2008. A comparative study of the therapeutic effects of the Zataria multiflora vaginal cream and metronidazole vaginal gel on bacterial vaginosis. *Phytomedicine***15**(12): 1025-1031.
12. Hashempur, MH, Hashempour, MM, Mosavat, SH, Heydari, M. 2017. Rhazes—his life and contributions to the field of dermatology. *JAMA dermatology***153**(1): 70-70.
13. Shakeri, A, Hashempur, M, Mojibian, M, Aliasl, F, Bioos, S, Nejatbakhsh, F. 2018. A comparative study of ranitidine and quince (*Cydonia oblonga* mill) sauce on gastroesophageal reflux disease (GERD) in pregnancy: a randomised, open-label, active-controlled clinical trial. *Journal of obstetrics and gynaecology***In press**: 1-7.
14. Shīrāzī, MA. 2009. *Makhzan al-advīyah (The Storehouse of Medicaments)*. Tehran University of Medical Sciences,: Tehran.
15. Ghasemi Dehkordi, N, Sajadi, S, Ghanadi, A, Amanzadeh, M, Asghari GhR, AG. 2002. Iranian herbal pharmacopoeia. *Ministry of Health Pub. Tehran***1**: 105.
16. Sajed, H, Sahebkar, A, Iranshahi, M. 2013. Zataria multiflora Boiss.(Shirazi thyme)—an ancient condiment with modern pharmaceutical uses. *Journal of ethnopharmacology***145**(3): 686-698.
17. Matthews DR, H, Rudenski AS, N, Treacher DF, T. 1985. Homeostasis model assessment: insulin resistance and beta cell function from fasting plasma glucose and insulin concentrations in man. *Diabetologia***28**(7): 412-419.
18. Rahimlou, M, Yari, Z, Hekmatdoost, A, Alavian, SM, Keshavarz, SA. 2016. Ginger supplementation in Nonalcoholic Fatty Liver Disease: A Randomized, Double-Blind, Placebo-Controlled Pilot Study. *Hepat Mon* **16**(1):e34897
19. de la Monte, SM, Pang, M, Chaudhry, R, Duan, K, Longato, L, Carter, J, et al. 2011. Peroxisome proliferator-activated receptor agonist treatment of alcohol-induced hepatic insulin resistance. *Hepatology Research***41**(4): 386-398.
20. Lee, C-Y, Lee, C-H, Tsai, S, Huang, C-T, Wu, M-T, Tai, S-Y, et al. 2009. Association between serum leptin and adiponectin levels with risk of insulin resistance and impaired glucose tolerance in non-diabetic women. *The Kaohsiung journal of medical sciences***25**(3): 116-125.
21. Gholamhoseinian, A, Fallah, H, Sharifi-far, F, Mirtajaddini, M. 2008. The inhibitory effect of some Iranian plants extracts on the alpha glucosidase. *Iranian journal of basic medical sciences***11**(1): 1-9.
22. Shim, Y-J, Doo, H-K, Ahn, S-Y, Kim, Y-S, Seong, J-K, Park, I-S, et al. 2003. Inhibitory effect of aqueous extract from the gall of *Rhus chinensis* on alpha-glucosidase activity and postprandial blood glucose. *Journal of ethnopharmacology***85**(2-3): 283-287.

23. Gholamhoseinian, A, Fallah, H, Sharififar, F. 2009. Anti-hyperglycemic activity of four plants extracts effective against alpha glucosidase in normal and diabetic rats. *Journal of Kerman University of Medical Sciences***16**:35-44
24. Shokrzadeh, M, Chabra, A, Ahmadi, A, Naghshvar, F, Habibi, E, Salehi, F, et al. 2015. Hepatoprotective effects of *Zataria multiflora* ethanolic extract on liver toxicity induced by cyclophosphamide in mice. *Drug Res***65**(4): 169-175.
25. Sakhaee, E, Emadi, L, Azari, O, Khanaman, FS. 2011. Evaluation of the beneficial effects of *Zataria multiflora* Boiss in halothane-induced hepatotoxicity in rats. *Advances in Clinical and Experimental Medicine***20**: 23-29.
26. Ong, JP, Pitts, A, Younossi, ZM. 2008. Increased overall mortality and liver-related mortality in non-alcoholic fatty liver disease. *Journal of hepatology***49**(4): 608-612.
27. Ryoo, JH, Suh, YJ, Shin, HC, Cho, YK, Choi, JM, Park, SK. 2014. Clinical association between non-alcoholic fatty liver disease and the development of hypertension. *Journal of gastroenterology and hepatology***29**(11): 1926-1931.
28. Monsour Jr, HP, Frenette, CT, Wyne, K. 2012. Fatty liver: a link to Cardiovascular disease—its natural history, Pathogenesis, and treatment. *Methodist DeBakey cardiovascular journal***8**(3): 21-25
29. Kensara, OA, ElSawy, NA, El-Shemi, AG, Header, EA. 2013. *Thymus vulgaris* supplementation attenuates blood pressure and aorta damage in hypertensive rats. *Journal of Medicinal Plants Research***7**(11): 669-676.
30. Aguila, MB, Silva, SPS, Pinheiro, AR, Mandarim-de-Lacerda, CA. 2004. Effects of long-term intake of edible oils on hypertension and myocardial and aortic remodelling in spontaneously hypertensive rats. *Journal of hypertension***22**(5): 921-929.
31. Gruenwald, J, Healthcare, T, Brendler, T, Jaenicke, C. 2007. *PDR for Herbal Medicines*. Thomson.